Libft 2-Week Mastery Roadmap

1337 / 42 School Curriculum — Structured Execution Plan

Target Duration: 14 Days | **Total Functions: 34+** | **Focus: Memory & Logic**

TIMELINE	FOCUS MODULE	TARGET FUNCTIONS / TOPICS	KEY LEARNING OBJECTIVE
WEEK 1: CHARACTER LOGIC, RAW MEMORY & STRINGS			
Days 1 – 2	**Character Checks & Conversions**	`ft_isalpha` `ft_isdigit` `ft_isalnum` `ft_isascii` `ft_isprint` `ft_strlen` `ft_toupper` `ft_tolower`	Master ASCII ranges, condition checks, size_t, and const pointers.
Days 3 – 4	**Raw Memory Byte Manipulation**	`ft_memset` `ft_bzero` `ft_memcpy` `ft_memmove` `ft_memchr` `ft_memcmp`	Understand generic void* pointers, casting to unsigned char*, and byte iteration.
Days 5 – 6	**String Searching & Copying**	`ft_strchr` `ft_strrchr` `ft_strncmp` `ft_strlcpy` `ft_strlcat` `ft_strnstr` `ft_atoi`	Pointer arithmetic, buffer bounds handling, and null-termination protection.
Day 7	**Review & Refactoring**	`Norminette Check, Edge Cases, Unit Testing`	Ensure zero memory leaks, code cleanliness, and full Norm compliance.
WEEK 2: DYNAMIC ALLOCATION, ADVANCED UTILITIES & BONUS			
Days 8 – 9	**Dynamic Memory Allocation**	`ft_calloc` `ft_strdup` `ft_substr` `ft_strjoin` `ft_strtrim`	Heap management with malloc(), overflow handling, and NULL-check guards.
Days 10 – 11	**Advanced Utilities & File Descriptors**	`ft_split` `ft_itoa` `ft_strmapi` `ft_striteri` `ft_putchar_fd` `ft_putstr_fd` `ft_putendl_fd` `ft_putnbr_fd`	2D arrays allocation/ freeing, function pointers, write() with file descriptors.
Days 12 – 13	**Bonus: Linked Lists & Makefile**	`ft_lstnew` `ft_lstadd_front` `ft_lstsize`	Self-referential structs, node traversal/cleanup, and Makefile rule automation.

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		`ft_lstlast` `ft_lstadd_back` `ft_lstdelone` `ft_lstclear` `ft_lstiter` `ft_lstmap`	
Day 14	**Final Audit & Peer Evaluation**	`Libft Testers, Valgrind, Peer Reviews`	Validation readiness: 125/100 target score.